Exact stability for nearly extremal chordal clique partitions

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Abstract

Let $M(n) = \lfloor n(n+1)/6 \rfloor$, and let $d = M(n) - \mathrm{cp}(G)$ be the deficit of a chordal graph. We combine the structural theorem of Paper IV with the parametric local formula of Paper VI. For sufficiently large order and $0 \le d \le n/2000$, the vertices of degree greater than $n/2$ form a clique P. Relative to this clique, the deficit is exactly the sum of the loss caused by its size, the number of missing cross edges, and twice the number of exterior edges. In particular, the graph differs from the associated complete split graph in at most d edges, and an optimal partition uses cliques of order at most four. The proof first obtains a coarse edit bound from IV, uses it to enter the local regime of VI, and then recovers the exact formula. This is an application of the two papers, with a separate formal companion.

A clique partition covers every edge exactly once by cliques of order at least two. Its minimum number of blocks is denoted by $\mathrm{cp}(G)$. We write $\mathrm{cp}_{\le 4}(G)$ when blocks are restricted to order at most four. All graphs are finite and simple. For disjoint vertex sets P, O, the notation $K_P \vee I_O$ denotes the complete split graph with clique P and independent set O.

The distinction between a local formula and a structural theorem matters here. Paper VI gives exact attainment once the graph is locally close to a suitable complete split graph. Paper IV supplies that reference clique for a nearly extremal chordal graph. The present note records how the two statements combine and keeps the resulting deficit identity explicit. The eventual Erdős 81 bound is already established in IV and in the other developments compared in its Section 8. This note concerns the stated stability consequence, not a new solution of that problem.

1. Exact stability in a linear deficit window

Put

$$M(n) = \left\lfloor \frac{n(n+1)}{6} \right\rfloor, \qquad B(n,p) = p(n-p) - \binom{p}{2}, \qquad \eta = 10^{-16}.$$

Theorem 1.1. Using the uniform far-rounding and near-structure theorems of Paper IV stated below, there exists an integer N such that every chordal graph G of order $n \ge N$, with deficit

$$d = M(n) - \mathrm{cp}(G), \qquad 0 \le d \le n/2000,$$

has the following description. The set

$$P = \{v : 2d_G(v) > n\}$$

is a clique. Write $p = |P|$, $O = V(G) \setminus P$, let A count missing edges between P and O, and put $m = e(G[O])$. Then

$$\boxed{\mathrm{cp}(G) = \mathrm{cp}_{\leq 4}(G) = B(n,p) - A - 2m,} \tag{1}$$

$$\boxed{d = \big(M(n) - B(n,p)\big) + A + 2m,} \tag{2}$$

$$\boxed{|E(G) \triangle E(K_P \vee I_O)| = A + m \leq d.} \tag{3}$$

In addition,

$$(6p - 2n - 1)^2 \leq 24d + 25. \tag{4}$$

Equation (2) separates the deficit into quantities that can be read from the graph once the clique is known. The first term measures the effect of its size; the other terms count actual missing cross edges and exterior edges. Equation (3) then bounds the total number of edits to the associated complete split graph. The degree test identifies the same clique without requiring a search over vertex bipartitions.

The size p need not be a maximizing size, such as $\lfloor (n+1)/3 \rfloor$. This is why the parametric Proposition 2.1 of VI, rather than its balanced Theorem A alone, is needed.

2. Structural and rounding inputs

Let $W^*(G)$ be the fractional maximum gain of a packing of triangles and copies of K_4, with gains 2 and 5. Write $F_4(G) = e(G) - W^*(G)$.

1. **Uniform rounding [1, Theorem 3.1].** For $\zeta = \eta/2$, once $n \geq N(\eta/2)$, there is a physical packing whose completion into cliques has at most

   $$F_4(G) + (\eta/2)n^2$$

   blocks. This input applies uniformly to the graphs in question, not only to a preselected fractional optimizer.
2. **Near-structure witness [1, Proposition 4.3, Theorem 5.0 and Lemma 5.3].** If $n \geq 4 \cdot 10^{12}$, G is chordal and

   $$F_4(G) \geq n^2/6 - \eta n^2,$$

   IV produces a literal clique root P, a physical packing and its completed partition of size L, with literal exterior-edge and missing-spoke counts m, A, satisfying

   $$L \leq B(n,p) - m/20 - A/2. \tag{5}$$

In the first input, completing a packing means covering every unused edge individually. The resulting object is a partition of the original graph, so its number of blocks is an upper bound for $\mathrm{cp}(G)$. In the second input, the root, packing, partition and counts belong to a single witness. We will use its partition to control the counts, and its root to apply VI.

The counts in this witness are the actual exterior edges and missing cross edges of its regularized clique root. Equation (5.15) of IV gives (5); its stronger form (5.15a) concerns the same witness. This physical identification is needed before the local theorem of VI can be applied. The separate formal companion records the source interfaces.

3. The order threshold

Choose N so that, for every $n \geq N$,

$$n \geq N(\eta/2), \qquad n \geq 4 \cdot 10^{12}, \qquad n \geq 100, \qquad \frac{\eta}{2} n^2 > \frac{n}{2000} + 1. \tag{6}$$

Such an N exists because $\eta > 0$. For the purely numerical last condition, $n \geq 2 \cdot 10^{13}$ is already sufficient; the unknown uniform-rounding threshold may be larger. This is an eventual theorem, not an all-orders numerical assertion.

4. Proof

We first exclude the far regime. We then use the near witness to bound the number of edited edges and the size of its clique root. Those two bounds verify the hypotheses of VI; only after applying its exact formula do we improve the edit bound and identify the root by degrees.

Suppose first that $F_4(G) < n^2/6 - \eta n^2$. Uniform rounding gives a physical clique partition with fewer than

$$\frac{n^2}{6} - \frac{\eta}{2} n^2 < \frac{n^2}{6} - \frac{n}{2000} - 1 \leq M(n) - d = \mathrm{cp}(G)$$

blocks, a contradiction. The last weak inequality uses $M(n) \geq n^2/6 - 1$ and $d \leq n/2000$. Thus the near-structure input applies.

Use its root P, its actual counts m, A and its partition of size L. The square identity gives $B(n,p) \leq Q(n) = (2n+1)^2/24$. For integer orders, $Q(n) - M(n)$ is $1/24$ when $n \equiv 0, 2 \pmod 3$, and $9/24$ when $n \equiv 1 \pmod 3$. Thus $Q(n) < M(n) + 1$; since $B(n,p)$ is an integer, $B(n,p) \leq M(n)$. Hence

$$M(n) - d = \mathrm{cp}(G) \leq L \leq B(n,p) - m/20 - A/2 \leq M(n) - m/20 - A/2.$$

The first two conclusions below follow by comparing the left and right sides of this chain. For the third, multiply the bound $m/20 + A/2 \leq d$ by 20 and use $A \geq 0$:

$$B(n,p) \geq M(n) - d, \qquad m/20 + A/2 \leq d, \qquad m + A \leq 20d. \tag{7}$$

Because this root is a clique, the complete split graph $S = K_P \vee I_O$ differs from G in exactly $m + A$ edges. Its maximum modification degree therefore satisfies

$$h = \Delta(G \triangle S) \leq m + A \leq 20d \leq n/100. \tag{8}$$

To control the root size, set $Q(n) = (2n+1)^2/24$. Completing the square gives

$$Q(n) - B(n,p) = \frac{(6p - 2n - 1)^2}{24}, \qquad 0 < Q(n) - M(n) < 25/24.$$

The square identity measures the distance of the root size from the center of the split parabola. By (7), $M(n) - B(n,p) \leq d$; adding the floor error $Q(n) - M(n)$ and multiplying by 24 proves (4). The bound on that floor error follows from $Q(n) = n(n+1)/6 + 1/24$ and the defining property of the floor. Since $d \leq n/2000$ and $n \geq 100$,

$$24d + 25 \le 0.012n + 25 \le n^2/100.$$

Taking square roots gives $|6p - 2n - 1| \le n/10$. Thus $1.9n + 1 \le 6p \le 2.1n + 1$. For $n \ge 100$, these inequalities imply the convenient coarse bounds

$$3n/10 \le p \le 2n/5. \tag{9}$$

Put $q = n - p$. Equations (8)–(9) yield

$$q - p = n - 2p \ge n/5 \ge 18h, \qquad 4p - q = 5p - n \ge n/2 \ge 36h.$$

These are precisely

$$p + 18h \le q \le 4p - 36h,$$

the hypotheses of VI Proposition 2.1 with its parameter $t_{\mathrm{par}} = 4$. Indeed, its inequalities are $p + (4t_{\mathrm{par}} + 2)h \le q$ and $q + t_{\mathrm{par}}(t_{\mathrm{par}} + 5)h \le t_{\mathrm{par}}p$. The required positivity of the core and the condition $q \ge p$ follow from (9) and the first inequality. The root has no missing internal edges. The local formula is therefore $pq - \binom{p}{2} - A - 2m$, which is (1). Subtracting it from $M(n)$ gives (2). All three summands in (2) are nonnegative, so $A + m \le A + 2m \le d$, proving (3). Notice that the initial bound $20d$ was only an entry budget; exact attainment improves it to d.

It remains to check that the root just used is the set specified in the theorem. A vertex of the clique has degree $n - 1$ in the complete split reference and loses at most h incident edges. An exterior vertex has reference degree p and gains at most h. Thus, for $a \in P$ and $x \in O$, the modification-degree bounds give

$$d_G(a) \ge n - 1 - h \ge 0.99n - 1 > n/2, \qquad d_G(x) \le p + h \le 0.41n < n/2.$$

Thus this same root is exactly the degree-defined set in the statement. This proves the theorem.

5. Relation to the series

The existence of the clique root and the exclusion of the far regime come from Paper IV. The exact formula in the resulting neighborhood comes from Paper VI. The initial bound $A + m \le 20d$ pays for entry into that neighborhood; exact attainment then improves it to $A + m \le d$. The abstract transfer inequality of Paper V does not supply the structural root. Its application to the eventual Erdős 81 bound is developed in the separate IV–V note.

When $d = 0$, equation (2) forces $A = m = 0$ and $B(n, p) = M(n)$. Thus the graph is complete split, with a maximizing clique size. That classification is already established in IV [1, Corollary 6.2] and Okechukwu [4, Corollary 1.2]. The present note records an exact deficit identity and degree recognition throughout the stated linear window.

Okechukwu [4, Theorem 1.3] proves qualitative stability for graphs with sublinear rooted simplicial defect; Section 7 asks for a quantitative relation between the partition deficit and edit distance. Paper IV [1, Theorem 6.1] already provides a linear edit bound to an associated complete split graph and distinguishes that graph from one with an optimizing core size. The present application combines IV with VI to obtain the coefficient-one bound (3) and the exact identity (2) in the smaller window $d \le n/2000$. Its edit distance is to the degree-identified graph

$K_P \vee I_O$, whose core size need not maximize $B(n,p)$. It therefore does not assert an optimal quantitative rate for the entire class considered in [4]. The original chordal clique-partition problem is due to Erdős, Ordman and Zalcstein [5].

Formalization and reproducibility

The separate Lean 4.35 project proves Theorem 1.1 from the proved inputs of IV and VI. The formal companion identifies its final source freeze, the public endpoint and physical adapters, the canonical root build and axiom audits, and the pinned reproduction commands. Its selected dependency chain excludes IV's final global-bound theorem while retaining IV's structural and rounding constructors. The source and audit records distinguish this project's eight modules from the exact upstream source cuts. Sources and formal companion.

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Use of artificial intelligence and computational tools

In the source papers used here, Claude, by Anthropic, and ChatGPT/Codex, by OpenAI, were used to explore and check arguments and to prepare the manuscripts. Aristotle, by Harmonic, contributed to exploring approaches, searching for counterexamples, developing proofs, and formalizing and reviewing them in Lean; its role was not limited to translating finished proofs.

For this note, ChatGPT/Codex assisted in assembling and checking the argument, developing the Lean interfaces, and preparing the manuscript and audit materials. Aristotle contributed a separate formalization proposal.

Certo and Jacobian were the computational tools used in the underlying research program to explore instances, check identities, and assess candidates, as disclosed in Paper IV. Their use is acknowledged separately from the logical dependencies of the formal proof. The finite regression checks performed specifically for this note use the exact-arithmetic scripts identified in the accompanying audit. Computations were reviewed within their stated scope; a finite search does not replace a universal proof.

The author remains responsible for the arguments, citations, code, and presentation. No AI system is listed as an author.

References

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[4] O. Okechukwu, *Clique partitions and bounded simplicial defect*, arXiv:2609.20871v1 (2026). https://arxiv.org/abs/2609.20871v1.

[5] P. Erdős, E. T. Ordman, and Y. Zalcstein, "Clique partitions of chordal graphs," *Combinatorics, Probability and Computing* **2** (1993), 409–415. https://doi.org/10.1017/S0963548300000808.